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DT/NT : DT

LESSON : AWS

SUBJECT : SNAPSHOTS-AMI

BATCH : B 424-020DE

AWS-DEVOPS

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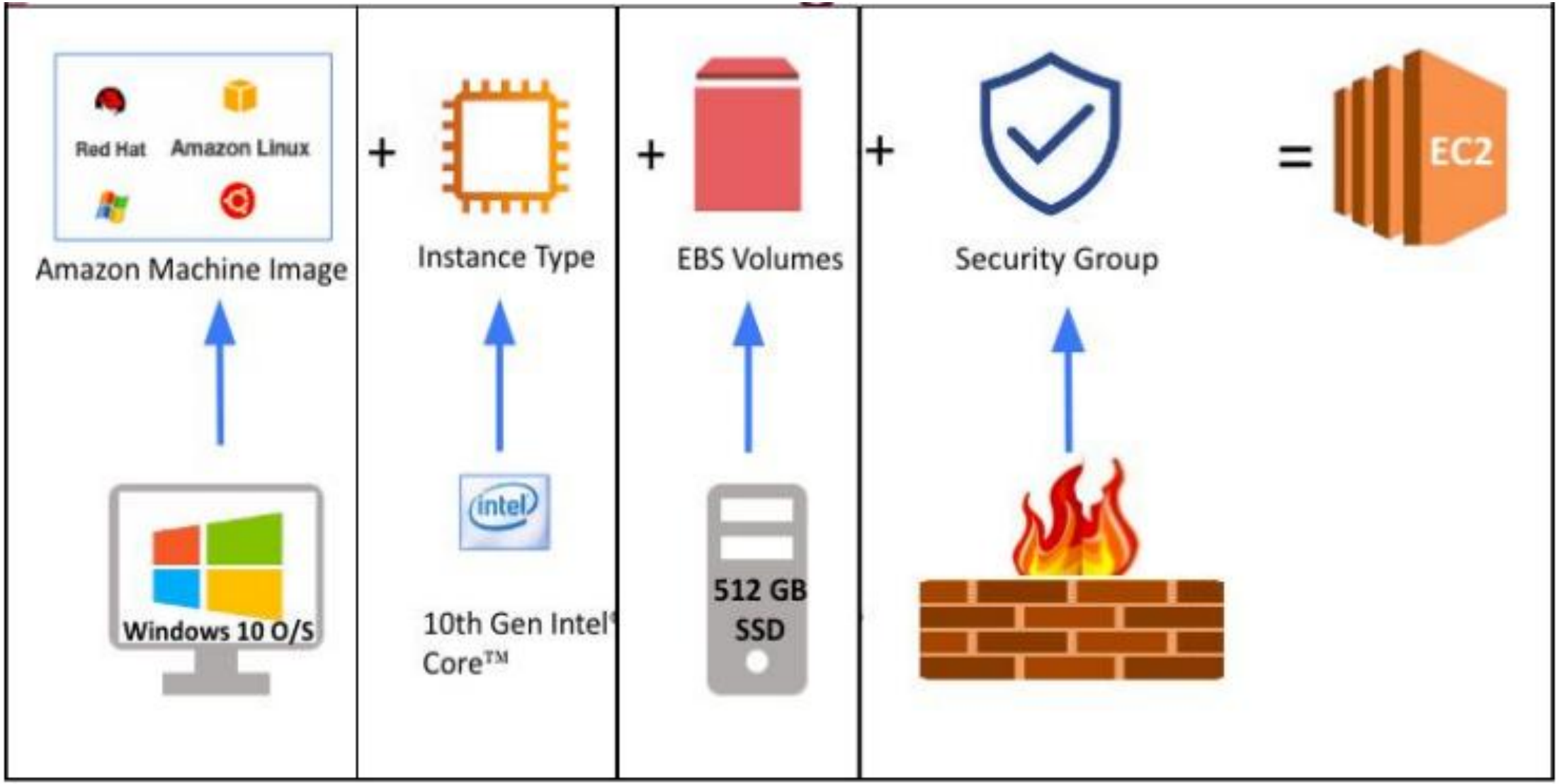
- Amazon Machine Images (AMIs)
- EBS Snapshots and Recovery
- Amazon Data Lifecycle Manager
- EC2 Hands-On and Cleanup

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Amazon Machine Image (AMI)

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Amazon Machine Image (AMI)



Amazon Machine Image (AMI)



Amazon Machine Image (AMI)

What is AMI?

- An AMI is a template used to launch an Amazon EC2 instance.

It includes root-volume content and launch information, such as architecture and block device mappings.

Launch multiple EC2 instances from the same AMI.

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Amazon Machine Image (AMI)

Types of AMIs

AMI visibility and distribution are different properties

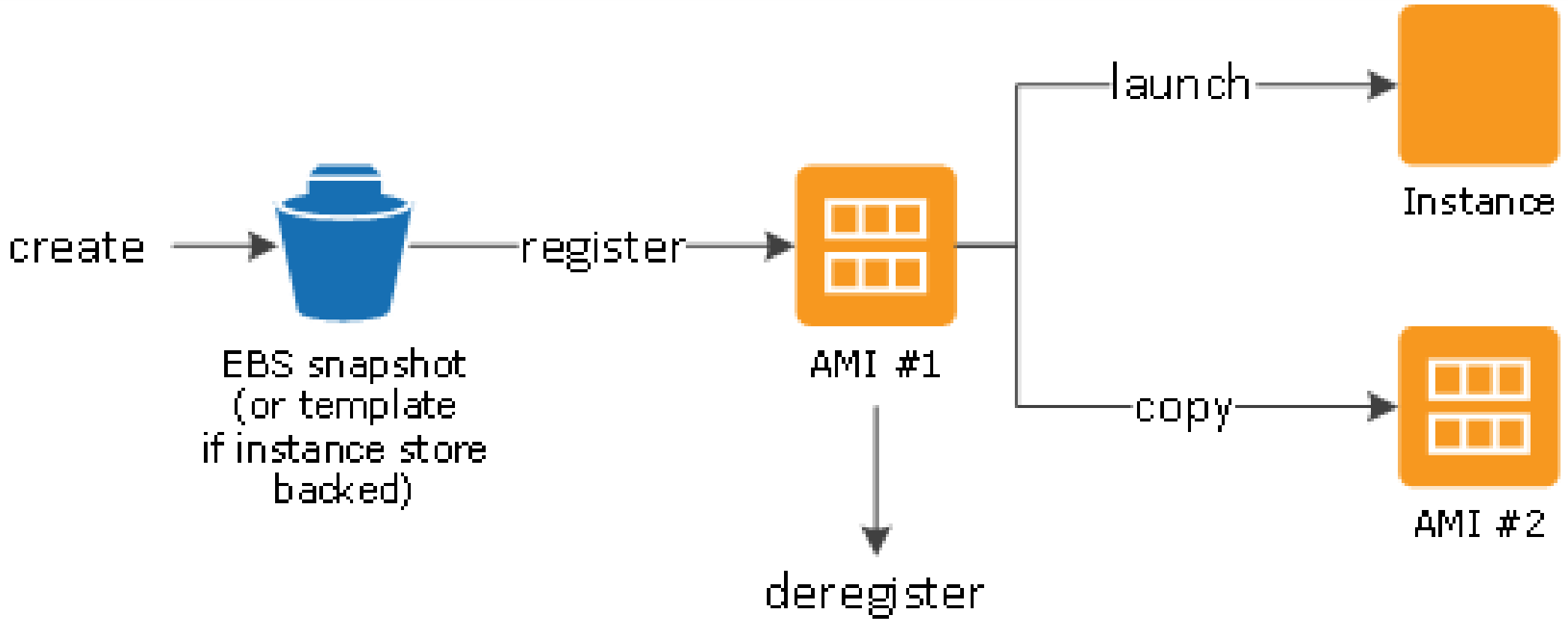
Private AMI	Public AMI	Marketplace AMI
Available to your account; share with selected accounts.	Can be discovered by other AWS accounts, if allowed.	Published by a seller; software charges may apply.

Always verify AMI owner, architecture, Region and licensing before launch.

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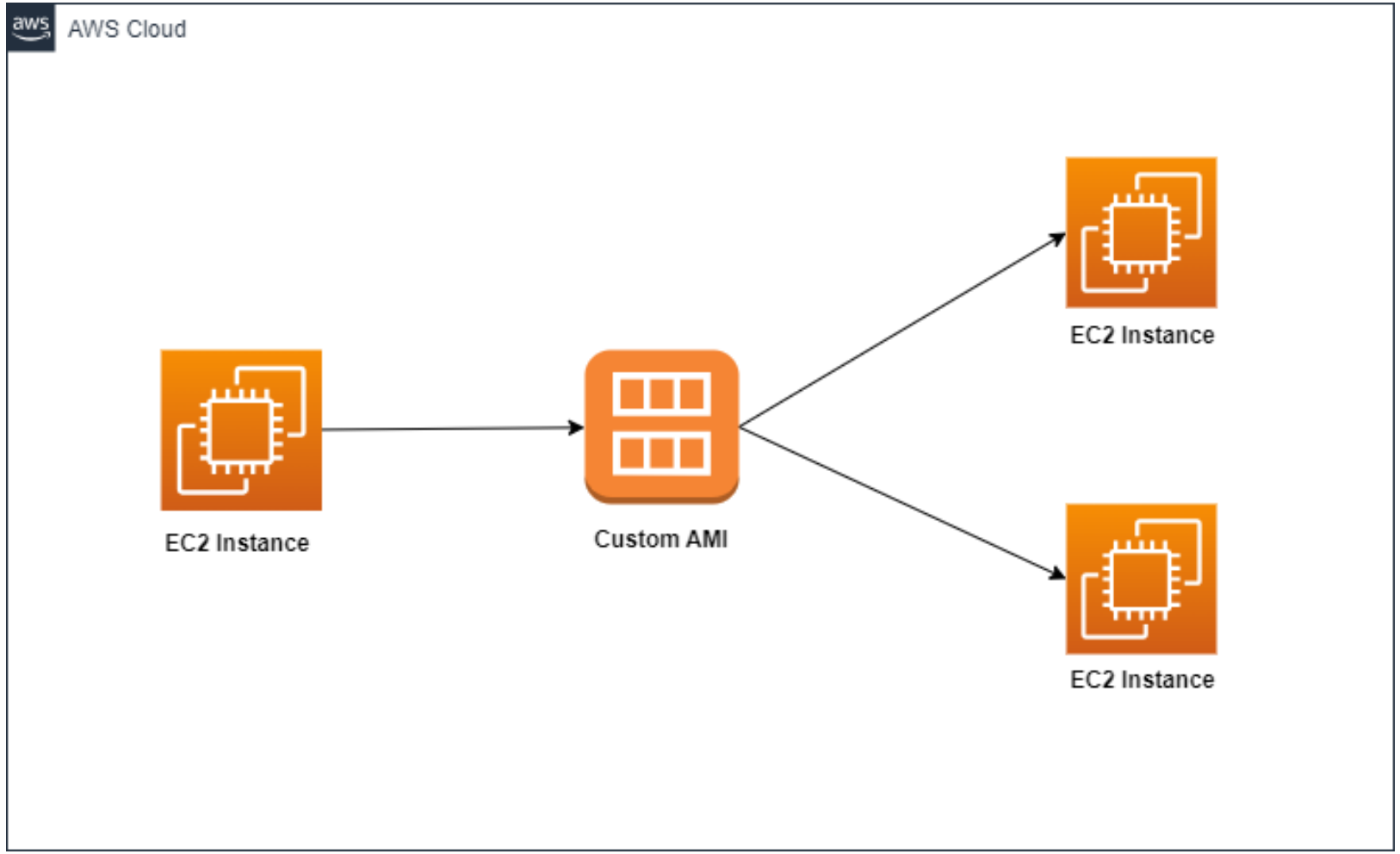
Amazon Machine Image (AMI)

AMI lifecycle



Amazon Machine Image (AMI)

AMI lifecycle



Snapshot

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Snapshot

What is Snapshot?



- An EBS snapshot is a point-in-time backup of an Amazon EBS volume.

Snapshots help you restore data or create a new volume.

Later snapshots store changed blocks incrementally.

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Snapshot



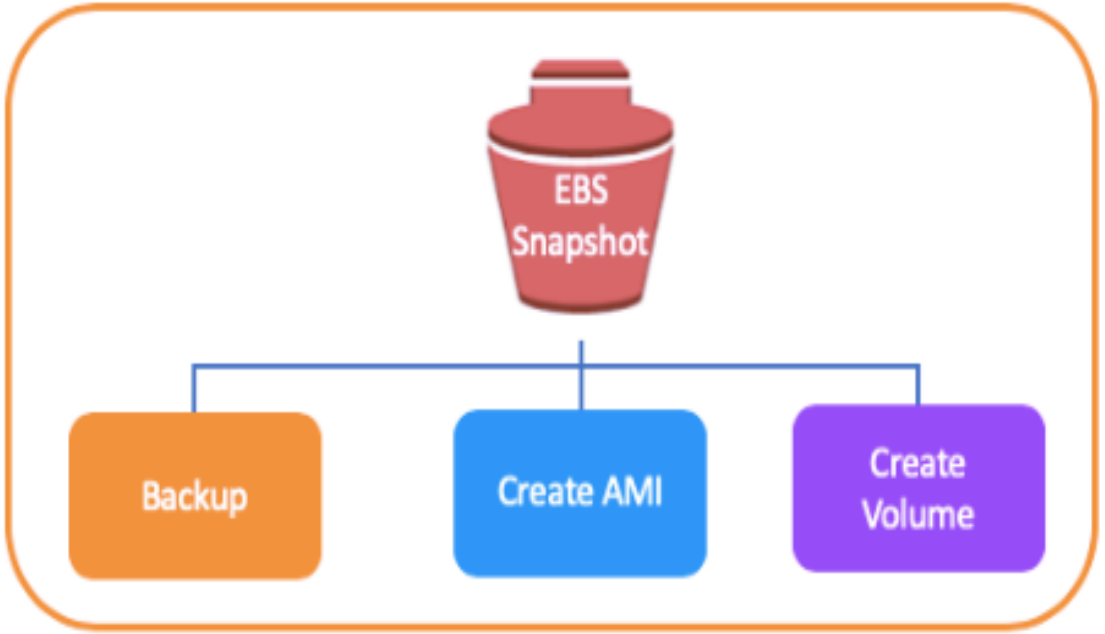
Snapshots are used for the purpose of

- Back up EBS volumes

Restore data to a new EBS volume

Copy backups to another AWS Region

Support EBS-backed AMIs



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Snapshot



Features of the Snapshot

- Created from EBS volumes (or an EC2 instance snapshot set)

Stored in AWS-managed Amazon S3 storage

Incremental blocks; not visible in your S3 buckets

Can be automated with Amazon DLM or AWS Backup

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Snapshot



Features of the Snapshot

- Copy an EBS snapshot to another AWS Region.

Create a new EBS volume from a snapshot in the destination Region.

An EBS volume belongs to one Availability Zone; volume-copy capabilities also exist.

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Snapshot

EBS snapshot and restore flow



EBS volume	EBS snapshot	New EBS volume
Original data on EC2	Point-in-time backup	Restore into selected AZ

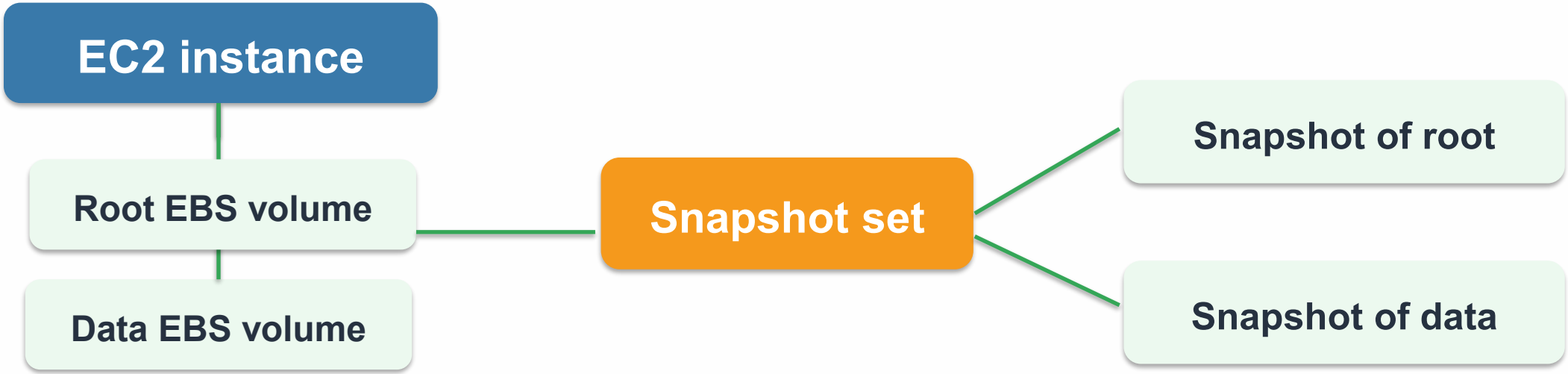
A snapshot is not a running EC2 instance or a mountable file system.

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Snapshot



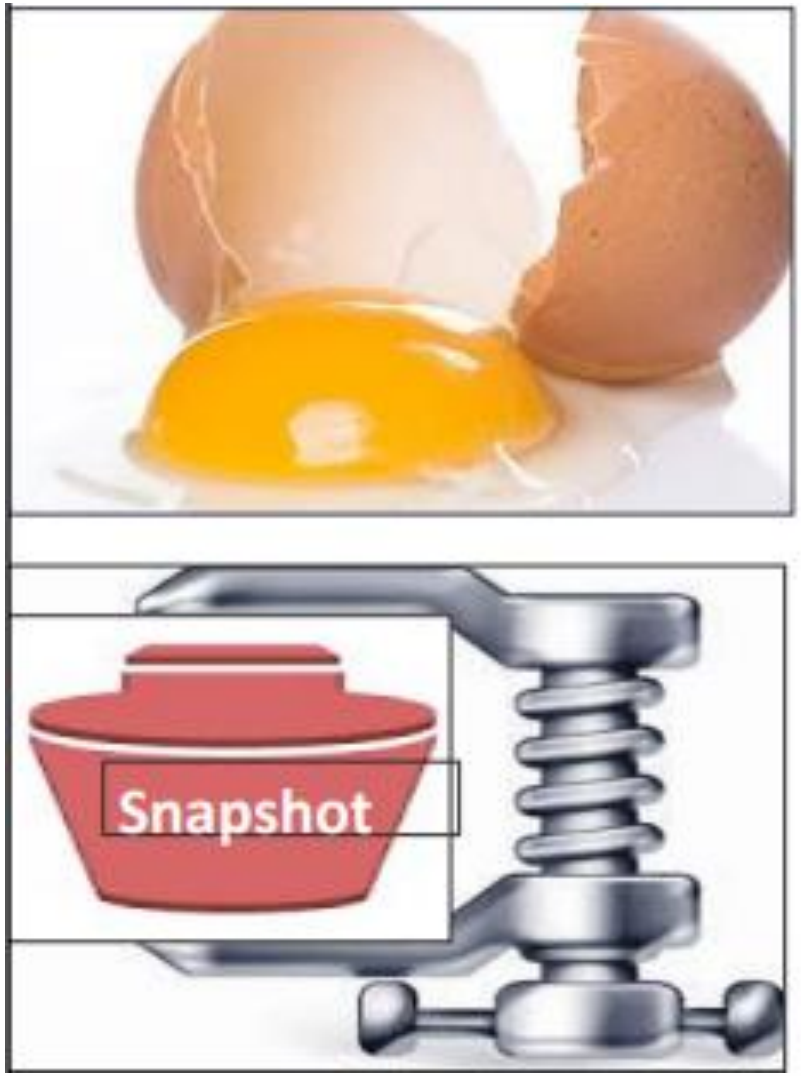
Multi-volume snapshots of an instance



A multi-volume snapshot creates an individual snapshot for each included volume.

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Snapshot

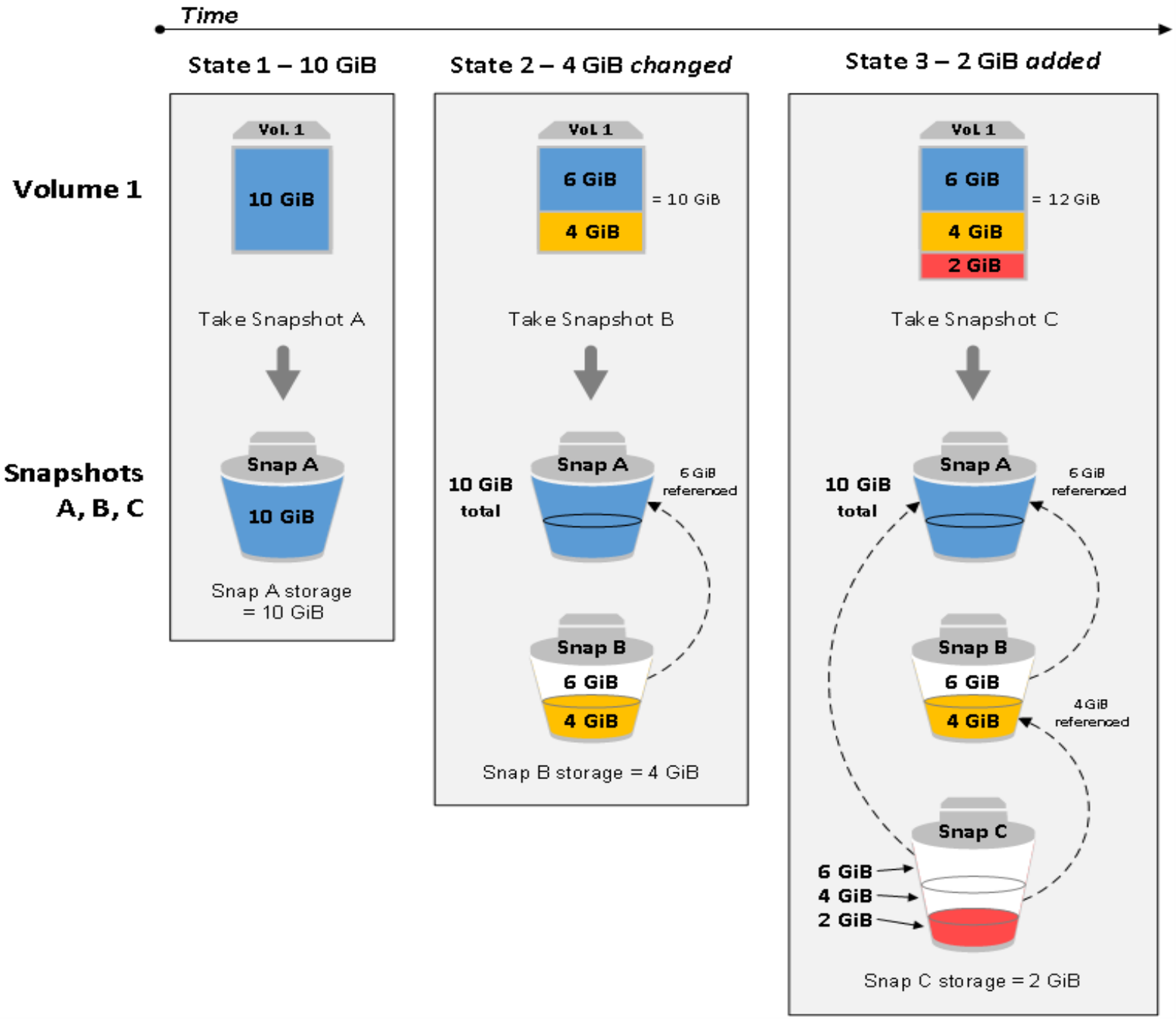


VS



Snapshot

Incremental Backups



Snapshot

Data Lifecycle Manager (Amazon DLM)

- Amazon Data Lifecycle Manager (DLM) automates EBS snapshot and EBS-backed AMI schedules.

Use tags to select resources, and configure retention to manage old backups.

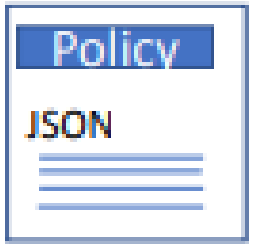
DLM manages resources created by its own policies.

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Snapshot

Data Lifecycle Manager (Amazon DLM)



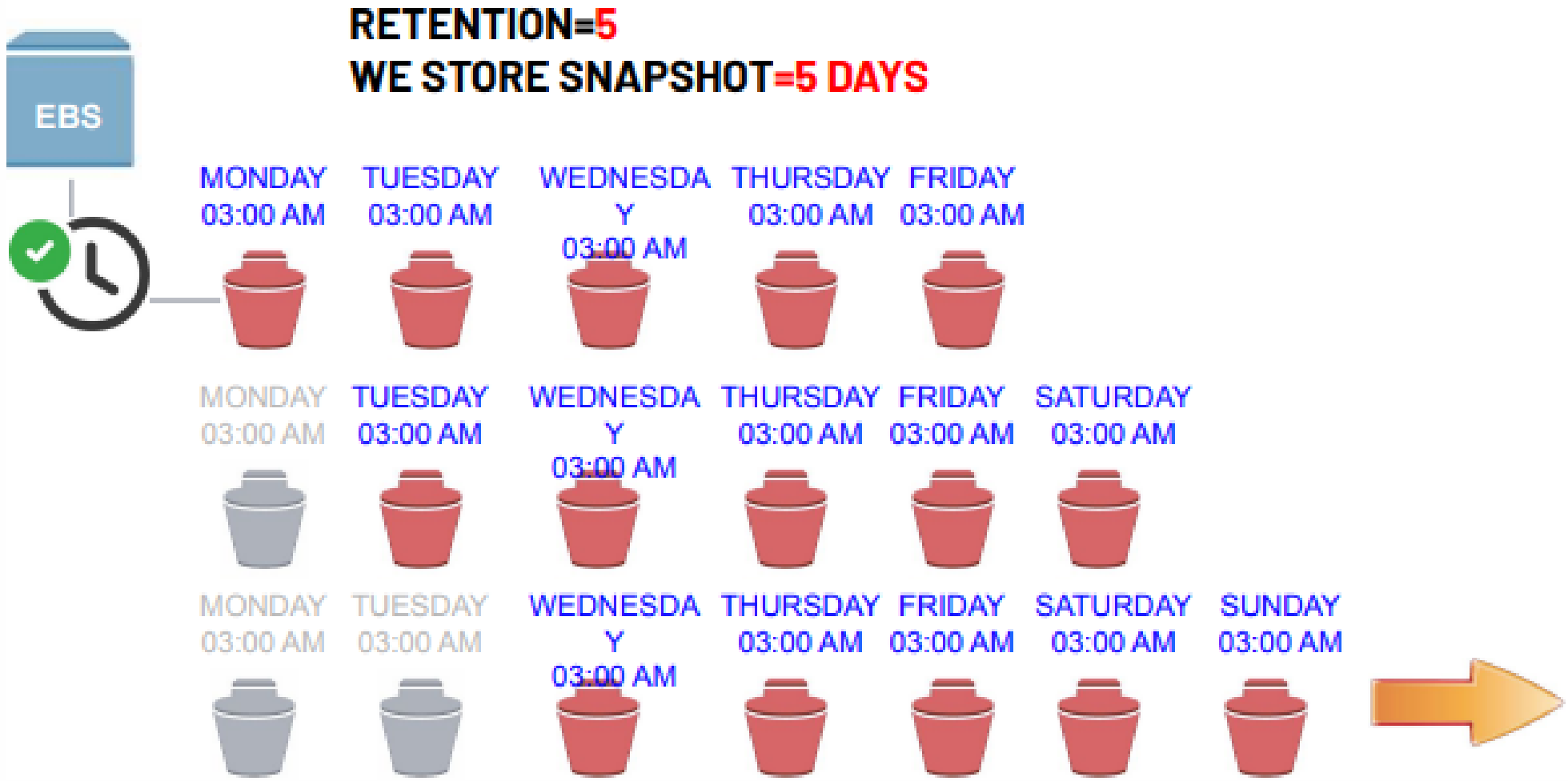


RETENTION=5

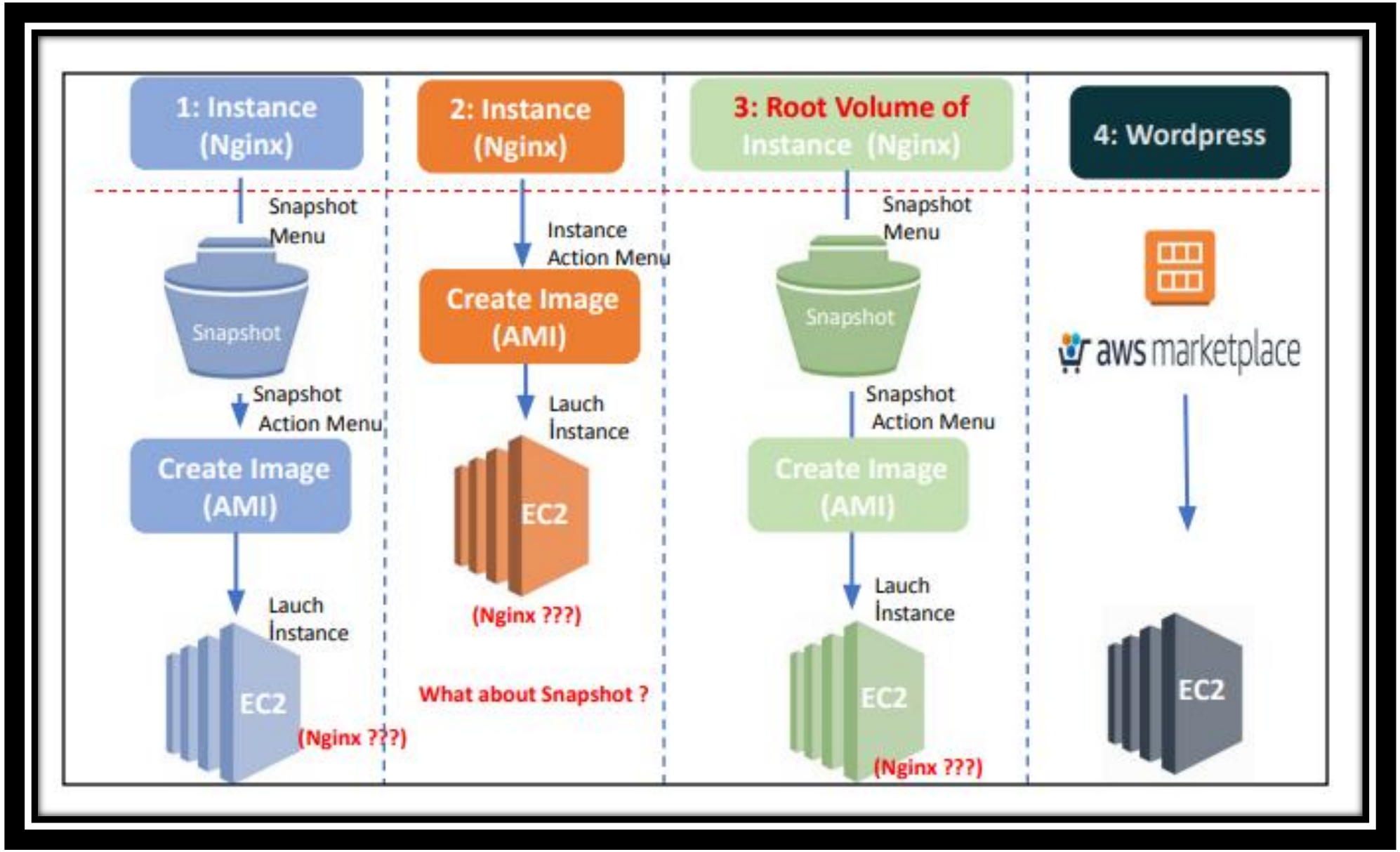
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Snapshot

Data Lifecycle Manager (Amazon DLM)



HANDS-ON


1: Instance (Nginx)
2: Instance (Nginx)
3: Root Volume of Instance (Nginx)
4: Wordpress
Snapshot Menu
Snapshot
Snapshot Action Menu
Create Image (AMI)
Lauch Instance
EC2
(Nginx ???)
Instance Action Menu
Create Image (AMI)
Lauch Instance
EC2
(Nginx ???)
What about Snapshot ?
Snapshot Menu
Snapshot
Snapshot Action Menu
Create Image (AMI)
Lauch Instance
EC2
(Nginx ???)
aws marketplace
EC2

HANDS-ON

AMI and Snapshot Lab Roadmap

1 — Build	Launch Amazon Linux 2023 + NGINX
2 — Back up	Create snapshot of the root EBS volume
3 — Reuse	Create an AMI and launch a new EC2 instance
4 — Automate	Create a tag-based DLM snapshot policy

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Snapshot vs. AMI

Choose the right recovery tool

EBS Snapshot

Backs up data on an EBS volume

Restore to a NEW EBS volume

Best for block-data recovery

Amazon Machine Image

Template for launching EC2

Includes launch metadata + root disk image

Best for cloning a configured server

AMI creation usually relies on EBS snapshots for EBS-backed instances.

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AMI Sharing

Share privately — do not publish classroom AMIs

Default	Keep your AMI private in the account.
Private sharing	Grant launch permission to a known AWS account ID.
Encryption	Encrypted snapshots may require KMS key permissions.
Public access	Keep block-public-access protection enabled in labs.

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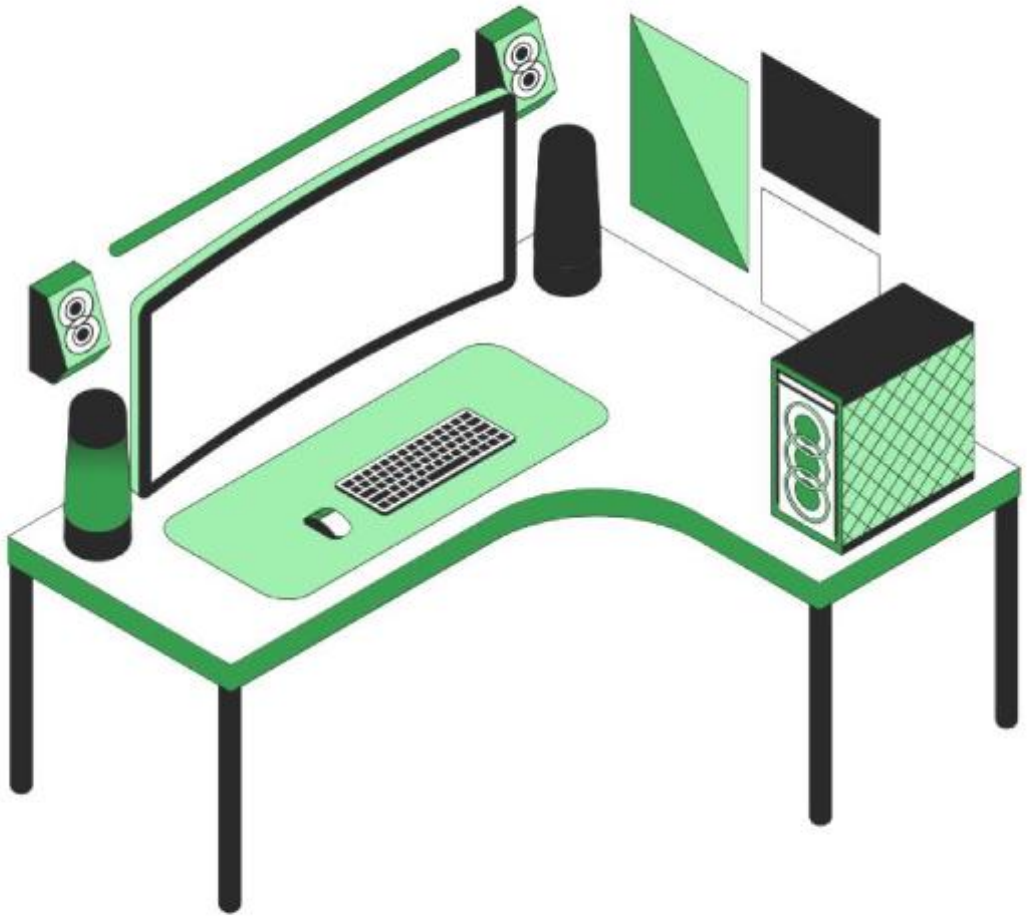
Finish the Lab Safely

Validation, costs, and cleanup

Validate	AMI status = Available; snapshot status = Completed.
Restore	Launch from the AMI; open the NGINX page.
Automate	Check DLM policy state and matching resource tags.
Clean up	Terminate EC2; deregister AMIs; delete unused snapshots/volumes.

Check billing: EBS, snapshots, public IPv4, and Marketplace licenses may incur charges.

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Do you have any questions?

Send it to us! We hope you learned something new.

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